SLC, causing independent clusters connected by a thin line of (noise) points to be mistakenly classified as a single cluster.

12.3.1 Directly vs Density-Reachable

To understand how DBSCAN works, we need to start from a few definitions; please use figure 12.9 as a reference while going through them, to help or check your understanding.

- Point `p`, in figure 12.9, is said to be a core point because there are at least `minPoints` points (including `p` itself) within distance ε from it (where, in the example, `minPoints==3`).
- Point `q` is *directly reachable* from `p` because point `q` is within distance ε from `p`, that is a core point. A point can only be directly reachable from a core point.
- A point `w` is *reachable* (or, equivalently, *density-reachable*) from a core point (like `p`) through a path of core points `p=w1, ..., wn=w`, if each `wi+1` is directly reachable from `wi`. From the definition of direct-reachability above, it follows that all points in the path, except `w`, needs to be core points.
- Any two points that are density reachable from each other are, by definition, in the same cluster.
- If there is any point `r` that is not reachable from any other point in the dataset, then `r` (and all the points like `r`) is marked as an outlier (or, equivalently, as noise).